6. Optimizing the design

Part B due Mar 6, 2024 20:30 IST

The next step is to study the system's behavior at other forcing frequencies, as we want to minimize the largest amplitude of the building's oscillations that can occur in the range $\Omega \in [0.7, 1.3]$.

LTI Consumer (External resource) (1.0 / 1.0 points)

Sweeping through input frequencies

The script below is designed to compute and plot the swaying amplitude of the building over a range of driving frequencies. The swayin TMD will also be plotted for comparison. To complete the script, you must:

1. Define the variable omSweep to be a row vector containing 50 equally spaced points over the interval $[0.7, 1.3]$. (Hint: use MAT
2. Copy the relevant lines of code from the previous problem to solve the system using ODE45.

To test the script, use the parameters:
$$\begin{aligned} m_1 &= 1, & m_2 &= 0.05,\\ k_1 &= 1, & k_2 &= 0.02,\\ b_1 &= 0.001, & b_2 &= 0.02. \end{aligned}$$

Use the initial conditions $\mathbf{x}_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$,

and solve the system for times t on the interval [0,7000] using linspace with 35000 points. If your script runs correctly, you will see how of the building for each forcing frequency Ω in the range $[0.7, 1.3]$.

Script

Save Reset MATLAB Documentation (https://www.mathworks.com/help/)

```
load('Reference.mat')

%Define parameters
m1 = 1;
m2 = 0.05;
k1 = 1;
k2 = 0.02;
b1 = 0.001;
b2 = 0.02;
omSweep = linspace(0.7, 1.3, 50);

%Numerically solve DE at each forcing frequency
for i=1:length(omSweep)

    om = omSweep(i);

    %Copy code to solve DE here
    %Numerically solve DE
    x0 = [0;0;0;0];
    tvec = linspace(0,7000,35000);
    A = [0,0,1,0;0,0,0,1;-(k1+k2)/m1,k2/m1,-(b1+b2)/m1,b2/m1;k2/m2,-k2/m2,b2/m2,-b2/m2];
    [t,x] = ode45(@(t,x) A*x + [0;0;sin(om*t)/m1;0],tvec,x0);

    %Compute steady state solution and extract building oscillation amplitude
    lt = length(t);
    per = 2*pi/om;
    [~,idx] = min(abs(t-(t(end)-5*per)));
    ampBuilding(i) = max(x(idx:lt,1));

end

%Plot
plot(OmRef,ampRef,'r'); hold on;
plot(omSweep,ampBuilding,'k');
xlim([0.7,1.3]); ylabel('Max. Amp. of Building'); xlabel('$\Omega$');
legend('Without TMD','With TMD')

```

Run Script

Previous Assessment: All Tests Passed

Submit

Calculator

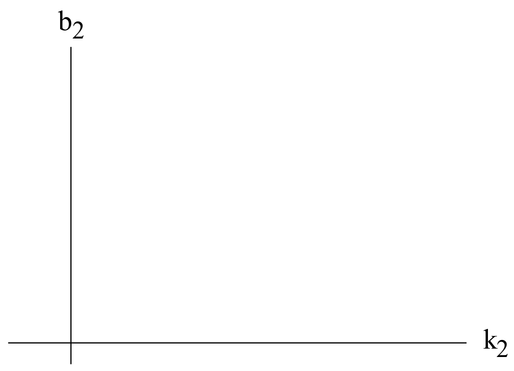
Correct definition of OmSweep

Correct numerical result

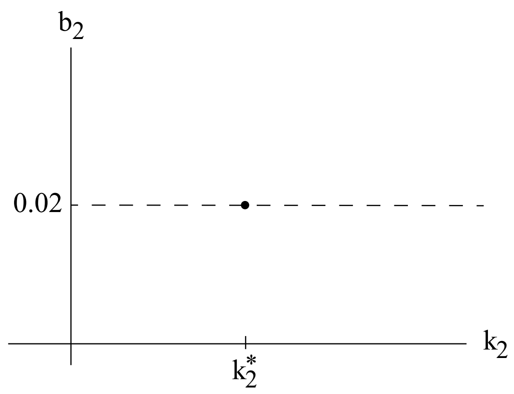
Optimization problem

The optimal values of k_2 and b_2 will minimize the maximum value of the black response curve.

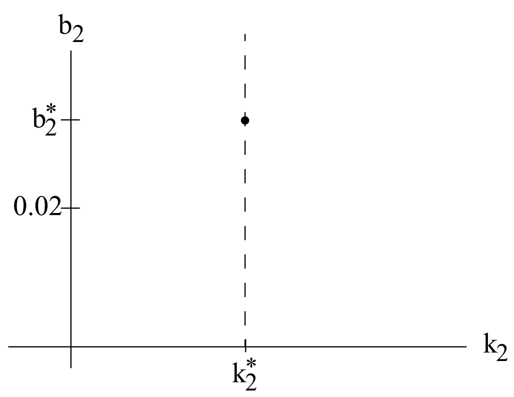
This is a two-dimensional minimization problem. We have a plane of possible values for k_2 and b_2 that we must attempt to minimize this problem over.



However, in our example where the building has a small damping constant, b_1, it can be shown that the parameters k_2 and b_2 can be optimized independently of each other. Therefore, our approach is to fix $b_2 = 0.02$ and first determine the optimal value, k_2^*, of the spring constant.



Next we fix $k_2 = k_2^*$ at this optimal value and determine the optimal value, b_2^*. (You can test that if you then fix b_2^*, and try to re-optimize k_2, your answer does not change.)



MATLAB exploration 1

1/1 point (graded)

Copy the script you used in the previous problem (*Sweeping through input frequencies*) to MATLAB online or on your computer. You will also need to save and copy the file ‘Reference.mat'. You can do this by typing

```
url = 'https://courses.edx.org/asset-v1:MITx+18.033x+1T2018+type@asset+block@Reference.mat';
websave('Reference.mat', url);
```

into your MATLAB command window.

While keeping all of the other parameters constant, change k_2 and see what effect it has on the response curve of the building. Based on your observations, enter the optimal value, k_2^*, that is accurate to three decimal places.

$k_2^* =$ 0.04492 ✔ Answer: 0.045

Calculator

$k_2^* =$ 0.04492 ✓ Answer: 0.045

Solution:

$k_2^* = 0.045$

Submit You have used 3 of 10 attempts

Answers are displayed within the problem

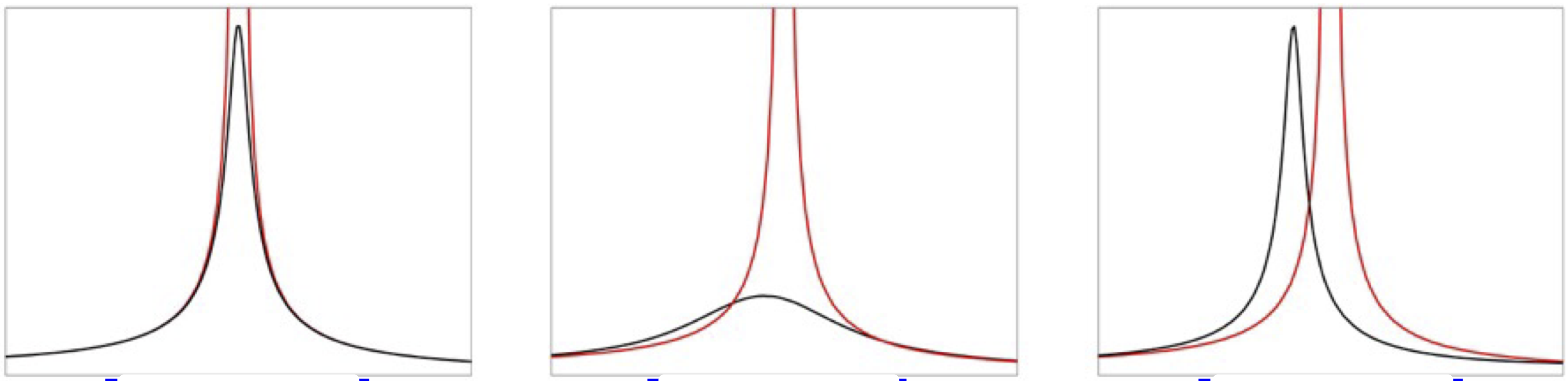
Matching graphs 1

3/3 point (graded)

Keyboard Help

PROBLEM

Match the three graphs to the corresponding inequalities based on your MATLAB exploration.



$k_2 < k_2^*$ | $k_2 = k_2^*$ | $k_2 > k_2^*$

Submit You have used 1 of 2 attempts.

Reset

Show Answer

FEEDBACK

✓ Correctly placed 3 items

✓ Your highest score is 3.0

Good work! You have completed this drag and drop problem.

MATLAB exploration 2

1/1 point (graded)

Now that you have found the optimal spring constant, k_2^*, the last step is to determine the optimal damping constant, b_2^*. Using your value of k_2^* and keeping all of the other parameters constant, vary b_2 and see what effect it has on the response curve of the building. Like in the previous problem, based on your observations, enter the optimal value, b_2^*, that is accurate to three decimal places.

$b_2^* =$ 0.013 ✓ Answer: 0.013

Solution:

$b_2^* = 0.013$

Submit You have used 2 of 10 attempts

Answers are displayed within the problem

Calculator

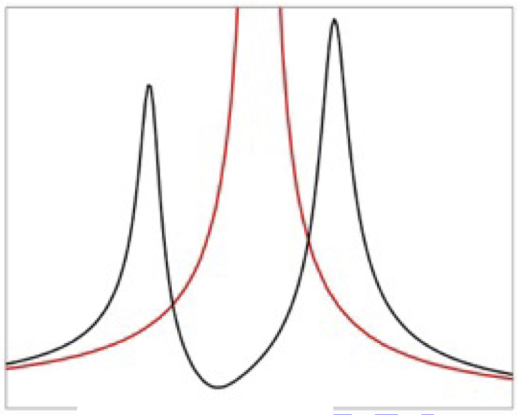
Matching graphs 2

3/3 point (graded)

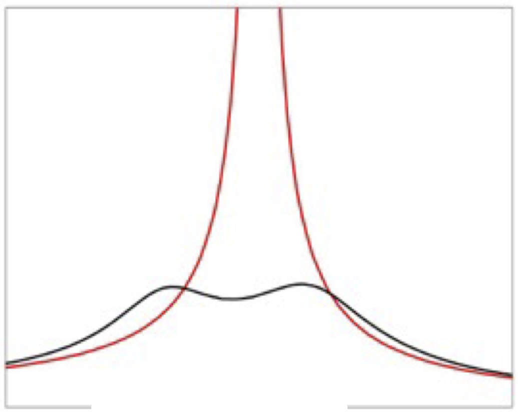
Keyboard Help

PROBLEM

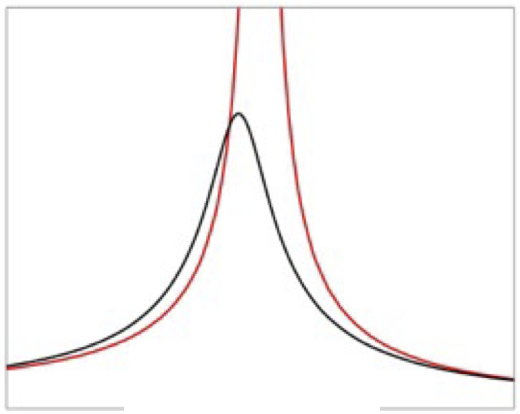
Match the three graphs to the corresponding inequalities based on your MATLAB exploration.



$b_2 < b_2^*$



$b_2 = b_2^*$



$b_2 > b_2^*$

Submit You have used 1 of 2 attempts.

Reset Show Answer

FEEDBACK

- Correctly placed 3 items
- Your highest score is 3.0
- **Good work! You have completed this drag and drop problem.**

We have achieved our goal and have determined the optimal spring and damping constants that minimize the swaying amplitude of the building over a range of forcing frequencies! You should double check that using the value of b_2 found in the previous step and then optimizing k_2 will give you the same result. (A general numerical approach to optimizing two parameters is to iteratively optimize each parameter independently, and using the new value to re-optimize the other parameter until no changes are detected.)

Note that this problem is a toy model. In reality, there are often many other factors that engineers must consider when designing a TMD for a building. For example:

- The building might sway in multiple dimensions and might also twist around its vertical axis.
- The building might need to withstand forcing from the ground due to seismic activity, which can produce a very different response compared to wind forcing.

6. Optimizing the design

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? STAFF - Part B Homework 3 (6. Optimizing the design) - #2 questions not visualized
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